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EduOS Cameroon — Costed Budget and Five-Year Total Cost of Ownership

Document ID	EDUOS-BUD-001
Version	1.0
Currency	FCFA (XAF), with USD equivalents at 574 FCFA/USD (Jul 2026); XAF is EUR-pegged
Costing basis	Bottom-up unit costing; every line's assumptions stated in §5
Status	Pre-appraisal estimate, class 3 (−20%/+30%)

1. Summary

Five-year programme total: 11.62 billion FCFA ≈ US$20.2 million, phased as below. The final programme year (Phase IV, Y5) costs 1.28 bn FCFA; from Year 6 the steady-state recurrent cost is **≈ 1.22 bn FCFA/year (US$2.1M)** (§4), equal to **0.13% of the two ministries' combined 2026 budgets** (929.5 bn FCFA) — the affordability test for national budget absorption after donor financing ends.

Phase	Years	Scope	Cost (bn FCFA)	US$M
I — Foundation & pilot	Y1–Y2	Core platform (registries, NTR, warehouse, school ops), 2 regions / 500-school pilot	4.44	7.7
II — Regional expansion	Y2–Y3	5 regions, ~8,000 public schools	3.15	5.5
III — National rollout	Y3–Y4	All 10 regions, ~18,000 public schools + private-school onboarding	2.75	4.8
IV — Consolidation (first year of full operation)	Y5	Full operation, optimization, module extensions	1.28	2.2
Total (5 years)			**11.62**	**20.2**

Scale basis (Baseline Data Annex §2): ~22,000 primary (14,000 public) + est. 4,500 secondary schools; 7.3M learners; 10 regions / 58 divisions / 360 sub-divisions. Rollout targets **public schools first** (~18,000–18,500), with private schools onboarding at marginal cost in Phase III (they use their own devices).

2. Budget by category (5-year)

#	Category	bn FCFA	US$M	%
1	Platform development (build, QA, security)	2.87	5.00	25%
2	Hosting infrastructure & DR	0.92	1.60	8%
3	School & office devices (two tiers incl. rugged tablets, solar kits, replacement)	2.54	4.43	22%
4	Training & capacity building	1.78	3.10	15%
5	Data cleaning & migration campaign	0.46	0.80	4%
6	Change management & communications	0.52	0.90	4%
7	Programme Management Unit & national tech team	1.61	2.80	14%
8	Independent verification, audit, M&E	0.29	0.50	2%
9	Contingency (after absorbing the rugged-tier premium, ADR-11)	0.63	1.09	5%
	Total	**11.62**	**20.22**	100%

3. Category detail and unit costs

3.1 PLATFORM DEVELOPMENT — 2.87 BN FCFA

Blended build team averaging 22 FTE over 30 months (national engineers majority, regional/international leads for architecture and security), fully-loaded blended rate 7.8M FCFA/FTE-month equivalent mix (≈US$13.6k blended — weighted 70% national at ~US$4–6k, 30% international at ~US$18–25k): 22 × 30 × 4.35M avg = 2.87 bn. Covers the Phase I module set (School/Student/Textbook registries, NTR per FRS EDUOS-FRS-NTR-001, warehouse, distribution, school operations, dashboards, sync engine), security audits (2 external penetration tests), and documentation. Later-phase module extensions are inside Phase III/IV shares of this line.

3.2 HOSTING & DR — 0.92 BN FCFA

National data-centre primary + DR site (RPO 24h/RTO 72h per Risk R16). Capex 0.35 bn (servers, storage, network for ~30M copy records / 300M events per FRS NFR-NTR-01); opex 0.115 bn/yr from Y2 (power, bandwidth, licences, admin). FCFA-denominated national hosting preferred (Risk R10).

3.3 DEVICES — 2.41 BN FCFA

Two device tiers, allocated per school from School Registry accessibility/connectivity/power data (ADR-11, doc 09):

Item	Qty	Unit (FCFA)	Total (bn)
Standard tier: Android 10+ smartphone/tablet, rugged case — connected schools	~13,000	92,000 (~US$160)	1.20
Rugged tier: MIL-STD-810H/IP68 tablet, ~22,000 mAh battery class (reference device: Blackview Active 8 Pro) — remote/off-grid schools operating travel-to-sync	~5,500	140,000 (~US$245)	0.77
Division/sub-division office kits (laptop + scanner; double as sync/charging points)	420	460,000	0.19
Solar charging kits — deepest-remote subset only (rugged tier's battery class covers weekly/monthly duty cycles between sync trips)	~3,500	69,000 (~US$120)	0.24
Replacement pool (15%/yr of school devices from Y3, Risk R9)	—	—	0.14 (within window)
Subtotal			**2.54**

The 0.13 bn increase over the single-tier baseline is absorbed by the contingency line (§2 line 9); final tier counts are outputs of the NSR data-cleaning campaign (§3.5), which therefore precedes device procurement.

3.4 TRAINING & CAPACITY BUILDING — 1.78 BN FCFA

Cascade: national trainers (60) → regional trainers (600) → school-level training (1 day, 2 staff per school, per-diem + materials ≈ 65,000 FCFA/school average incl. facilitation) × 18,500 schools = 1.20 bn; national training academy curriculum + refresher e-learning 0.23 bn; ministry/division officer training 0.35 bn. (Risk R3 mitigation.)

3.5 DATA CLEANING & MIGRATION — 0.46 BN FCFA

Division-level validation workshops for school lists and enrolment (58 divisions × 2 rounds), text-book catalogue and stock declarations import per FRS FR-NTR-MIG-01/02. (Risk R8 mitigation.)

3.6 CHANGE MANAGEMENT & COMMUNICATIONS — 0.52 BN FCFA

Stakeholder engagement (publishers, printers, unions, PTAs), national comms campaign, school-recognition programme (Risk R3).

3.7 PMU & NATIONAL TECHNICAL TEAM — 1.61 BN FCFA

PMU (coordinator, procurement, finance, safeguards, M&E, and 2 helpdesk posts operating the L2 support line from pilot go-live per MIP §5 track 3 — 8 staff × 5 yrs) 0.69 bn; national technical team ramping 4→14 engineers/analysts by Y4 (knowledge-transfer counterpart to vendor, Risk R4) 0.92 bn. From Y5, this team is the platform's permanent operator.

3.8 INDEPENDENT VERIFICATION — 0.29 BN FCFA

Annual technical + financial audit, third-party M&E data verification (aligns with M&E Framework EDUOS-MEF-001), and source-code escrow verification (Risk R4).

4. Recurrent cost after the programme (Year 6 onward)

Line	bn FCFA/yr
Hosting, licences, bandwidth, DR	0.115
National technical team (14 staff, operations + evolution)	0.28
Device replacement (15% of fleet)	0.28
Refresher training & onboarding of new staff	0.20
Connectivity stipends for school sync (average 1,500 FCFA/school/month where needed)	0.28
Audit & M&E	0.06
Total steady-state	**≈ 1.22 bn FCFA (US$2.1M)**

= **0.13% of combined MINEDUB+MINESEC 2026 budgets**, and ≈ 3.8% of the 2025 PAREC allocation alone. This line is proposed for inclusion in the medium-term expenditure framework from Y4 (Risk R5 mitigation).

5. Key costing assumptions (open for appraisal challenge)

1. Exchange rate 574 FCFA/USD; EUR-pegged XAF limits USD volatility to EUR/USD movement (Risk R10; 10% contingency held).
2. Public-school rollout = 18,500 sites; private schools onboard via web/BYOD at marginal cost.
3. One device per school is the floor; large schools (>1,500 learners) may need a second device — absorbed by the replacement pool in Y3–Y5 or a Phase IV top-up.
4. Blended development rate assumes majority-national team; a fully international build would roughly double line 1 — this is a deliberate sourcing-strategy choice, not padding.
5. QR labels are **not** in this budget: label printing at press is mandated in print contracts at ≈1–2% of unit book price (FRS §3.2, Risk R12) and sits in the existing textbook procurement budget, not the platform budget.
6. No school connectivity build-out is budgeted (that is a national telecom agenda); the design assumes offline-first with periodic sync (FRS §9).
7. Costs exclude e-content/digital textbook licensing (NEDIH module scope, separate future business case).

6. Financing strategy

Source	Target share	Rationale
IDA / World Bank (ERSP successor or additional financing)	50–60%	Direct continuity with P160926's textbook component; this platform protects that investment's returns
GPE system-capacity grant	15–20%	Systems-strengthening window fits registry/EMIS-type infrastructure
Government of Cameroon (PIB + recurrent from Y4)	20–25%	Ownership signal required by Risk R5; recurrent line into MTEF
Other partners (AFD, UNICEF, AfDB digital windows)	5–10%	Device/solar and training lines are cleanly earmarkable

Phase gates align with the Risk Register (§5): no phase disburses until the prior phase's M&E targets (EDUOS-MEF-001) are verified.